

Byzantine Reliable Broadcast in Wireless Networks[★]

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Abstract. Byzantine reliable broadcast (BRB) is a fundamental primitive in distributed computing. BRB ensures that messages can be reliably broadcast among nodes, regardless of the presence of Byzantine nodes. Research on BRB in wireless networks receives relatively less attention, while there have been significant advancements in wired networks. Although current works have yielded remarkable results, they still have some limitations. Specifically, the forwarding frequency of each node is $O(n)$, and the fault-tolerant thresholds still have a gap with the thresholds in wired networks where less than one-third of nodes can be faulty. In this paper, we propose a new BRB protocol that outperforms the state-of-the-art (SOTA) presented in PODC '05 in the following three respects:

- It improves the threshold in the L_∞ metric from $f < \frac{1}{2}r(2r+1)$ to $f < \lfloor \frac{1}{2}(r+1)(2r+1) \rfloor$.
- It improves the threshold in the L_2 metric from $f < 0.23\pi r^2$ to $f < \lfloor 0.3\pi r^2 \rfloor$.
- It reduces the forwarding frequency from $O(n)$ to $O(1)$.

In summary, our BRB has a constant forwarding frequency and achieves thresholds closer to the thresholds in wired networks. We provide the derivations of new thresholds and formally prove the correctness of our BRB protocol.

Keywords: Byzantine fault-tolerant · Reliable broadcast · Wireless networks.

1 Introduction

Byzantine reliable broadcast (BRB) is a fundamental primitive in distributed computing. The process begins with a source node that intends to disseminate messages to all other nodes in the system. Once a correct node reliably delivers a message, all other correct nodes will eventually reliably deliver the same message, even if some nodes (including the source node) are Byzantine. While noteworthy breakthroughs have been achieved, these endeavors predominantly concentrate on wired networks, overlooking the crucial aspect of wireless networks. However,

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BRB holds equal significance for applications in wireless (or radio) networks. We underscore two pivotal distinctions in achieving BRB in a wireless network, one being advantageous and the other detrimental:

- Each node has a broadcast radius r , and any message transmitted by a source node is received uniformly by all its neighbors (i.e., those within a distance of r from the source node). This property is advantageous for BRB because a node *cannot* equivocate to its neighbors.
- When two nodes are situated beyond a distance of r , their communication relies on the assistance of intermediary nodes for forwarding. Consequently, a source node colluding with an intermediary node retains the potential for equivocation.

These two characteristics highlight the unique opportunities and challenges in implementing BRB in wireless networks. Although the wireless channel is typically unreliable and prone to collisions, to better understand and address these aspects, starting with an idealized wireless channel is a practical approach. In an idealized wireless channel, the messages broadcast by a source node will eventually be received by its neighbors in the sent order, and no collision will occur. The results in [11,4,15,5,3] are also based on the same assumption.

Koo [11] proves that achieving BRB in an asynchronous wireless network is possible only when $f < \frac{1}{2}r(2r + 1)$ in the L_∞ metric¹ where f denotes the number of Byzantine nodes in any single neighborhood. Bhandari et al. [4] introduce a BRB protocol that aligns with this established threshold and shows that BRB is achievable when $f < 0.23\pi r^2$ in L_2 metric². Although these works achieve remarkable results, they still have some limitations. In these works, the forwarding frequency of each node is $O(n)$. The high forwarding frequency is costly in energy for wireless nodes, especially for energy-limited nodes (where n is the number of nodes within a neighborhood). Furthermore, the fault-tolerant thresholds still have a gap with the thresholds in wired networks where less than one-third of nodes can be faulty.

Our contribution. In this paper, we propose a new BRB protocol using the *aggregate signature* [6,14], which has $O(1)$ forwarding frequency and higher fault-tolerant thresholds than SOTA [4]. The aggregate signature is a cryptographic technique that allows multiple signatures from different nodes to be combined into a single signature, reducing the computational overhead required to verify signatures. In our BRB protocol, we use this technique to generate *certificates* that contain essential information enabling nodes to trust the content of messages, thus ensuring reliable message delivery. By using the aggregate signature technique, each node only needs to forward a certificate to convince others to reliably deliver the corresponding message, so our BRB achieves an $O(1)$ forwarding frequency. With low forwarding frequency demand, nodes do not need to have a large number of connections with others to receive and forward $O(n)$

¹ In the L_∞ metric the distance between two nodes $\mathcal{N}_{(x_1, y_1)}$ and $\mathcal{N}_{(x_2, y_2)}$ is $\max\{|x_1 - x_2|, |y_1 - y_2|\}$.

² In the L_2 metric, the distance is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

messages. This reduced connection requirement means that a node can have more faulty neighbors. As a result, our BRB is more resistant to node failures. Specifically, we improve the fault-tolerant threshold to $f < \lfloor \frac{1}{2}(2r+1)(r+1) \rfloor$ in the L_∞ metric. We also present arguments suggesting that in the L_2 metric, when r is sufficiently large and $f < \lfloor 0.3\pi r^2 \rfloor$, BRB can be achieved. Compared to the SOTA [4] that has fault-tolerant thresholds of $f < \frac{1}{2}r(2r+1)$ in the L_∞ metric and $f < 0.23\pi r^2$ in the L_2 metric, our thresholds are closer to the thresholds in wired networks where one-third of nodes can be faulty. Finally, we provide a detailed derivation of new thresholds and a formal correctness proof of our BRB.

Organization. In the remainder of this paper, we first provide some preliminaries of our BRB in Section 2. In Section 3, we provide details of our proposed BRB and the derivation of new thresholds. Then, we provided the proofs for our BRB in the L_∞ and L_2 metrics in Section 4. In the end, we discuss the related work in Section 5 and conclude this paper in Section 6.

2 Preliminaries

2.1 Model

Wireless grid network. In this paper, we focus on an infinite (or finite toroidal) wireless grid network [11] where nodes are located on a grid consisting of 1×1 square units. Each node can be uniquely identified by its grid location (x, y) , denoted as $\mathcal{N}_{(x,y)}$. All nodes have a broadcast radius of r , where $r \in \mathbb{Z}$. The messages broadcast by $\mathcal{N}_{(x,y)}$ can be received by all the nodes within a distance r from it. We refer to these nodes as the neighbors of $\mathcal{N}_{(x,y)}$. We remark that $\mathcal{N}_{(x,y)}$ and all its neighbors are termed as the *neighborhood* of (x, y) . For brevity, we may use $\mathcal{N}_i$ to denote $\mathcal{N}_{(x,y)}$ (i.e., use i to denote (x, y)). In this paper, we consider two distance metrics, namely L_∞ metric¹ and L_2 metric². The analysis under the L_∞ metric is more manageable, which facilitates us in determining precise fault-tolerant thresholds. Moreover, it offers insightful intuitions that serve as the basis for our approximate reasoning regarding the L_2 metric of practical significance.

Communication assumptions. We assume an asynchronous communication, where the sent message will eventually be received, but there is no time bound on the message delay. Following [11, 4, 5, 15, 3], we assume an *idealized shared wireless channel* where the message collision does not exist, and intrinsically, the messages broadcast by a source node will eventually be received by its neighbors in the sent order. Although real-world wireless channels are often unreliable and susceptible to collisions, it is beneficial to focus on an idealized wireless channel as a first step to investigate BRB. Specifically, this assumption allows us to first explore the fundamental principles and algorithms of BRB without the interference of complex real-world factors. Elements such as message conflicts and signal attenuation in real channels can then be gradually incorporated to refine and improve our proposed solutions, making them more practical and applicable.

Adversary. In this paper, we consider Byzantine faults; a Byzantine (or faulty) node can behave arbitrarily, but cannot cause collisions. Moreover, faulty nodes cannot forge a valid signature without access to the private key. We assume that up to f faulty nodes can exist in any neighborhood.

2.2 Problem Definition

Byzantine reliable broadcast. A source node can reliably broadcast a message m in round γ by calling $\text{r_bcast}(m, \gamma)$. A node can reliably deliver a message m in round γ from $\mathcal{N}_i$ by outputting $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$. The round number γ is used to differentiate the messages broadcast by a single node. Formally, a BRB protocol satisfies the following properties:

- **Validity:** if a correct node $\mathcal{N}_i$ calls $\text{r_bcast}(m, \gamma)$, then all correct nodes will eventually output $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$.
- **Agreement:** if a correct node outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$, then all other correct nodes will output $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$.
- **Integrity:** For each round γ and node $\mathcal{N}_i$, every correct node outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$ at most once regardless of m .

2.3 Aggregate Signature Scheme

In a naive approach to signing a message, each node has a public/secret pair of keys. All nodes know the public keys of others, while they only know their own secret key. Nodes use secret keys to sign a message and output a signature. A signature can be verified by using the corresponding public key. In other words, to validate a signed message, a node must know the sender's public key in advance. Therefore, in an infinite grid, we assume that a finite number of nodes can be the source node of reliable broadcast.

A certain number of signatures usually serve as a *certificate* to identify that a message is signed by some nodes. The *aggregate signature* scheme can compress multiple signatures of a message into a single aggregate signature. Each node can verify the aggregate signature via an aggregate public key. As a result, this scheme significantly reduces the space and computation needed for verification compared to the naive approach when multiple signatures are required as certificates. How to achieve it is out of the scope of this paper; in fact, we can implement it based on Schnorr [14] or BLS [6]. The key operations of the scheme we used are as follows:

- $\text{Sign}(\text{SK}_i, m)$: Input: a secret key SK_i and a message m .
Output: a signature σ of $\mathcal{N}_i$ for a message m .
- $\text{Aggre}(\sigma_1, \dots, \sigma_k)$: Input: a set of signatures $\sigma_1, \dots, \sigma_k$ of a common message.
Output: an aggregate signature $\hat{\sigma}$.
- $\text{Verify}(\text{PK}_i, m, \sigma)$: Input: a public key PK_i , and a signature σ of a message m .
Output: *true* when σ is $\mathcal{N}_i$'s signature of message m ; otherwise *false*.

In fact, the Aggre operation also outputs a bitmap that indicates the set of nodes $\hat{\mathcal{N}}$ that contribute to the aggregate signature $\hat{\sigma}$. This bitmap enables the operation Verify to compute the aggregate public key $\hat{\text{PK}}$ correspondingly. Nodes can use $\text{Verify}(\hat{\text{PK}}, m, \hat{\sigma})$ to check whether $\hat{\sigma}$ is a valid aggregate signature of nodes $\hat{\mathcal{N}}$ for m . Since the bitmap has no impact on the core conclusions in the subsequent analysis, for the sake of brevity, we omit the bitmap in subsequent sections. In the proposed BRB protocol, we use an aggregate signature to serve as a certificate to convince nodes to reliably deliver messages. More details are available in Section 3.

3 Proposed Byzantine Reliable Broadcast

In this section, we introduce our BRB protocol. It works when $f < \lfloor \frac{1}{2}(2r+1)(r+1) \rfloor$ in the L_∞ metric and when $f < \lfloor 0.3\pi r^2 \rfloor$ in the L_2 metric. We consider two types of nodes:

- *Nodes that are neighbors of the source node.* They can reliably deliver the message directly because the source node cannot equivocate with its neighbors.
- *Nodes that are not neighbors of the source node.* They reliably deliver a message upon receiving a valid *certificate* — that is, an aggregate signature contributed by $(f+1)$ nodes within the neighborhood of the source node. Since at most f nodes can be malicious in any neighborhood, a node receiving this certificate is convinced that at least one correct node has reliably delivered the corresponding message.

These two handling mechanisms cover all nodes and ensure the properties of our BRB protocol under the specified fault-tolerant thresholds in different metrics. Fig. 1 shows the communication pattern of our reliable broadcast:

1. Upon receiving a message from the source node, a neighbor of the source node reliably delivers the message and broadcasts a signed **committed** message (Fig. 1a).
2. Upon receiving a **committed** message from a neighbor of the source node, a node waits for $f+1$ **committed** messages from the neighbors of the source node (including itself), combines their signatures into a certificate (an aggregate signature) and broadcasts the certificate (Fig. 1b).
3. When receiving a valid certificate for the first time, a node reliably delivers the corresponding message (if it has not delivered any message in this round) and forwards the certificate to others (Fig. 1c).

As we can see, each node in the system only needs to broadcast a certificate message (neighbors of the source node also broadcast a **committed** message). Therefore, the forwarding frequency of each node is $O(1)$. The details of our BRB protocol are shown in Fig. 2. Note that each node only processes the messages with valid signatures. For brevity, we omit signature verification in Fig. 2.

Next, we introduce how $f < \lfloor \frac{1}{2}(2r+1)(r+1) \rfloor$ and $f < \lfloor 0.3\pi r^2 \rfloor$ are derived in the L_∞ and the L_2 metrics, respectively.

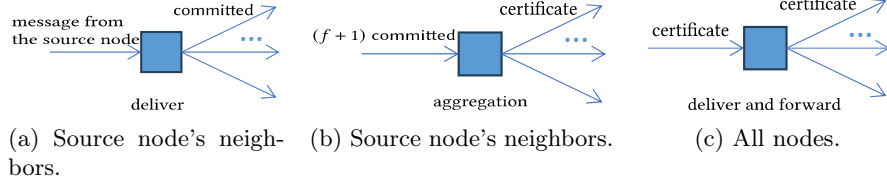


Fig. 1. The communication pattern of our reliable broadcast protocol.

Reliable Broadcast

By calling $\text{r_bcast}(m, \gamma)$, the source node $\mathcal{N}_s$ computes $\sigma_{\gamma,s} := \text{Sign}(\text{SK}_s, M)$ and broadcasts $\langle \text{committed}, s, m, \gamma, \sigma_{\gamma,s} \rangle$, where $M = \langle \text{committed}, s, m, \gamma \rangle$. Then, nodes run as the following:

1. Neighbors $\mathcal{N}_i$ of the source node $\mathcal{N}_s$.

- Upon receiving the first valid **committed** message from $\mathcal{N}_s$ for m in round γ :

$\mathcal{N}_i$ outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_s)$;

$\mathcal{N}_i$ computes $\sigma_{\gamma,i} := \text{Sign}(\text{SK}_i, M)$, where $M = \langle \text{committed}, s, m, \gamma \rangle$;

$\mathcal{N}_i$ broadcasts $\langle \text{committed}, s, m, \gamma, \sigma_{\gamma,i} \rangle$.

- Upon receiving $f+1$ valid **committed** messages for m in round γ :

$\mathcal{N}_i$ computes $\hat{\sigma}_{\gamma,m} := \text{Aggre}(\{\sigma_*\}_{f+1})$;

$\mathcal{N}_i$ broadcasts $\langle \text{certificate}, s, m, \gamma, \hat{\sigma}_{\gamma,m} \rangle$.

- When receiving a valid $\langle \text{certificate}, s, m, \gamma, \hat{\sigma}_{\gamma,m} \rangle$ for the first time:

$\mathcal{N}_i$ forwards this **certificate** message.

2. Non-Neighbors $\mathcal{N}_i$ of the source node $\mathcal{N}_s$.

- When receiving a valid $\langle \text{certificate}, s, m, \gamma, \hat{\sigma}_{\gamma,m} \rangle$ for the first time:

$\mathcal{N}_i$ forwards this **certificate** message;

$\mathcal{N}_i$ outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_s)$, if it has not delivered any message in round γ .

Fig. 2. The details of our Byzantine reliable broadcast protocol.

3.1 Derivation of $f < \lfloor \frac{1}{2}(2r+1)(r+1) \rfloor$ in the L_∞ metric

To better understand the derivation, we first explain some necessary information of Fig. 3. The source node is located at (a, b) with a broadcast radius r . The nodes $\mathcal{N}_{(i,j)}$, where $\{(i, j) | a-r \leq i \leq a+r; b-r \leq j \leq b+r\}$, within the blue region are neighbors of the source node $\mathcal{N}_{(a,b)}$. Node $\mathcal{N}_P$ is located at $(a+\beta r, b)$ where $0 \leq \beta \leq 1$. Nodes $\mathcal{N}_{(i,j)}$, where $\{(i, j) | |i-a| + |j-b| \leq \beta r\}$, within the yellow region in Fig. 3a are the more important nodes, which means that compared to $\mathcal{N}_P$, they have more (or the same number) shared neighbors with the source

node. The yellow region contains $2 \sum_{k=1}^{\beta r} (2k-1) + 2\beta r + 1 = (2r+1)((2-\beta)r+1)$ nodes. Nodes $\mathcal{N}_{(i,j)}$, where $\{(i,j) | a - (1-\beta)r \leq i \leq a+r; b-r \leq j \leq b+r\}$, within the dark green region in Fig. 3b are the shared neighbors of node $\mathcal{N}_P$ and the source node $\mathcal{N}_{(a,b)}$. The dark green region contains $(2r+1)((2-\beta)r+1)$ nodes. Nodes $\mathcal{N}_{(i,j)}$, where $\{(i,j) | a+1 \leq i \leq a+r; b-r \leq j \leq b+r\}$, within the red region in Fig. 3c are the neighbors on the right of the source node $\mathcal{N}_{(a,b)}$. The red region contains $(2r+1)r$ nodes. We summarize them in Table 1.

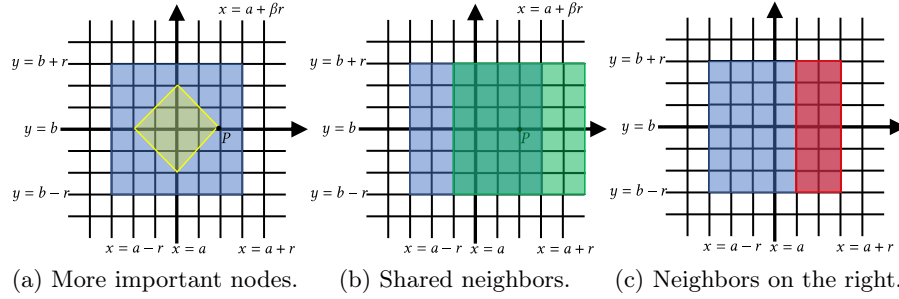


Fig. 3. Derivation of f in an infinite grid under L_∞ metric.

Table 1. The summarized necessary information of Fig. 3.

Region	x -extent and y -extent	Description of nodes	Numbers of nodes
Blue	$a-r \leq x \leq a+r$ $b-r \leq y \leq b+r$	The neighbors of $\mathcal{N}_{(a,b)}$	$(2r+1)^2$
Yellow	$ x-a + y-b \leq \beta r$	The more important nodes	$(2r+1)((2-\beta)r+1)$
Dark green	$a-(1-\beta)r \leq x \leq a+r$ $b-r \leq y \leq b+r$	The shard neighbors of $\mathcal{N}_P$ and $\mathcal{N}_{(a,b)}$	$2\beta r(\beta r+1) + 1$
Red	$a+1 \leq x \leq a+r$ $b-r \leq y \leq b+r$	The neighbors of $\mathcal{N}_{(a,b)}$ on the right	$(2r+1)r$

The intuition of this derivation is that “a valid **certificate** message will be constructed and received by all correct nodes”.

We first consider how the neighborhood of a source node ensures that “a valid **certificate** message will be constructed”. To construct a **certificate** message, a node must receive $f+1$ valid **committed** messages. Recall that only the neighbors of the source node broadcast **committed** messages. This implies that a node must have at least $2f+1$ shared neighbors with the source node because up to f nodes can be faulty. Obviously, compared to node $\mathcal{N}_P$, each node within the yellow region has more (or the same number) shared neighbors with the source node. This means that if node $\mathcal{N}_P$ can construct a valid **certificate** message, all nodes within the yellow region can also construct a valid **certificate** message. Therefore, to have at least one correct node construct a valid **certificate** message,

the number f of faulty nodes in the neighborhood of a source node should meet the following two rules:

1. The node $\mathcal{N}_P$ has at least $2f + 1$ shared neighbors with the source node: $(2r + 1)((2 - \beta)r + 1) \geq 2f + 1$.
2. The yellow region contains at least $f + 1$ nodes: $f < 2\beta r(\beta r + 1) + 1$.

Rule 1 ensures that all nodes within the yellow region can construct a valid **certificate** message. Rule 2 ensures that the yellow region contains at least one correct node. In summary, these two rules ensure that a valid **certificate** message can be constructed by at least one correct node.

Then, we consider how every neighborhood ensures that “a valid certificate will be received by all correct nodes”. In order to spread a **certificate** message to all directions (that is, up, down, left, and right), each node should have at least one correct node in each direction. As shown in Fig. 3c, all neighbors on the right of node $\mathcal{N}_{(a,b)}$ are within the red region, which contains $(2r + 1)r$ nodes. By symmetry, each node has $(2r + 1)r$ nodes in every direction. Therefore, one of these $(2r + 1)r$ nodes should be correct, which means that $f < (2r + 1)r$.

In summary, the number f of faulty nodes in each neighborhood should meet:

1. $(2r + 1)((2 - \beta)r + 1) \geq 2f + 1$
2. $f < 2\beta r(\beta r + 1) + 1$
3. $f < (2r + 1)r$

In fact, when $\beta = 0.8$, we get the maximum $f < \frac{1}{2}(2r + 1)(1.2r + 1)$. However, when $\beta = 0.8$, node $\mathcal{N}_P$ cannot be located at a grid point because $\beta r \bmod 1 \neq 0$. Thus, we make $\beta = 1$ conservatively to allow node $\mathcal{N}_P$ to be located at a grid point and get $f < \lfloor \frac{1}{2}(2r + 1)(r + 1) \rfloor$. In conclusion, each neighborhood in our protocol can tolerate $f < \lfloor \frac{1}{2}(2r + 1)(r + 1) \rfloor$ faulty nodes.

The L_∞ metric simplifies the analysis, enabling precise fault-tolerance thresholds. It also provides intuitive insights that underpin an approximate argument for the practically significant L_2 metric. Next, we introduce derivation of $f < \lfloor 0.3\pi r^2 \rfloor$ in the L_2 metric.

3.2 Derivation of $f < \lfloor 0.3\pi r^2 \rfloor$ in the L_2 metric

In this section, we introduce the derivation of $f < \lfloor 0.3\pi r^2 \rfloor$ in the L_2 metric. Note that we provide an approximate argument based on [4], for sufficiently large r , the area is a good approximation for the number of nodes in these regions (it is increasingly accurate for large r).

The intuition of this derivation is the same as that in the L_∞ metric, i.e., “a valid **certificate** message will be constructed, and received by all correct nodes”.

Similarly, to better understand this derivation, we first explain some necessary information of Fig. 4. The source node is located at (a, b) , and its broadcast radius is r . Nodes $\mathcal{N}_{(i,j)}$, where $\{(i, j) | (i - a)^2 + (j - b)^2 \leq r^2\}$, within the blue region are neighbors of the source node $\mathcal{N}_{(a,b)}$. Node $\mathcal{N}_P$ is located at $(a + \alpha r, b)$ where $0 \leq \alpha \leq 1$. Nodes $\mathcal{N}_{(i,j)}$, where $\{(i, j) | (i - a)^2 + (j - b)^2 \leq (\alpha r)^2\}$, within

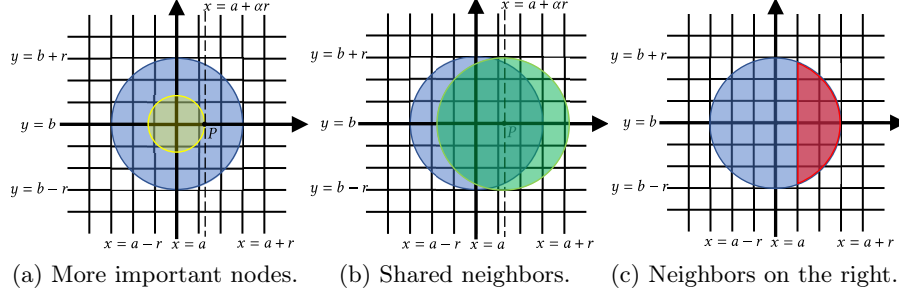


Fig. 4. Derivation of f in an infinite grid under L_2 metric.

Table 2. The summarized necessary information of Fig. 4.

Region	x -extent and y -extent	Description of nodes	Numbers of nodes
Blue	$(x - a)^2 + (y - b)^2 \leq r^2$	The neighbors of $\mathcal{N}_{(a,b)}$	πr^2
Yellow	$(x - a)^2 + (y - b)^2 \leq (\alpha r)^2$	The more important nodes	$\pi(\alpha r)^2$
Dark green	$(x - a)^2 + (y - b)^2 \leq r^2$ $(x - a)^2 + (y - b)^2 \leq (\alpha r)^2$	The shard neighbors of $\mathcal{N}_P$ and $\mathcal{N}_{(a,b)}$	$2r^2 \cos^{-1} \frac{\alpha}{2} - \frac{1}{2} r^2 \alpha \sqrt{4 - \alpha^2}$
Red	$(x - a)^2 + (y - b)^2 \leq r^2$ $a + 1 \leq x$	The neighbors of $\mathcal{N}_{(a,b)}$ on the right	$0.5\pi r^2 - 2r - 1$

the yellow region in Fig. 4a are the more important nodes, which means that compared to $\mathcal{N}_P$, they have more (or the same number) shared neighbors with the source node. The yellow region contains $\pi(\alpha r)^2$ nodes. Nodes $\mathcal{N}_{(i,j)}$, where $\{(i,j) | (i-a)^2 + (j-b)^2 \leq r^2; (i-a)^2 + (j-b)^2 \leq (\alpha r)^2\}$, within the dark green region in Fig. 3b are the shared neighbors of node $\mathcal{N}_P$ and the source node $\mathcal{N}_{(a,b)}$. The dark green region contains $2r^2 \cos^{-1} \frac{\alpha}{2} - \frac{1}{2} r^2 \alpha \sqrt{4 - \alpha^2}$ nodes. In more detail, let two circles of radius R and r and centered at $(0,0)$ and $(d,0)$ intersect in a region shaped like an asymmetric lens. The area of this lens is³:

$$r^2 \cos^{-1} \left(\frac{d^2 + r^2 - R^2}{2dr} \right) + R^2 \cos^{-1} \left(\frac{d^2 + R^2 - r^2}{2dR} \right) - \frac{1}{2} \sqrt{(-d+r+R)(d+r-R)(d-r+R)(d+r+R)}$$

Thus, we can calculate the area of the dark green region in Fig. 4b by making $R = r$ and $d = \alpha r$, and get the area:

$$2r^2 \cos^{-1} \frac{\alpha}{2} - \frac{1}{2} r^2 \alpha \sqrt{4 - \alpha^2}.$$

Nodes $\mathcal{N}_{(i,j)}$, where $\{(i,j) | (i-a)^2 + (j-b)^2 \leq r^2; a+1 \leq i\}$, within the red region in Fig. 4c are the neighbors on the right of the source node $\mathcal{N}_{(a,b)}$. The red region contains $0.5\pi r^2 - 2r - 1$ nodes. We summarize them in Table 2.

Similarly to the derivation in Section 3.1, the number f of the faulty nodes in each neighborhood should meet:

³ <https://mathworld.wolfram.com/Circle-CircleIntersection.html>

1. $2r^2 \cos^{-1} \frac{\alpha}{2} - \frac{1}{2}r^2 \alpha \sqrt{4 - \alpha^2} \geq 2f + 1$
2. $f < \pi(\alpha r)^2$
3. $f < 0.5\pi r^2 - 2r - 1$

When $\alpha = \sqrt{0.3}$, we get the maximum $f < 0.3\pi r^2$. Note that we do not select a strictly α to locate node $\mathcal{N}_P$ on a grid point because the threshold in this metric is an approximate argument. Therefore, in our protocol, each neighborhood should tolerate $f < \lfloor 0.3\pi r^2 \rfloor$ faulty nodes.

4 Correctness proof

4.1 Correctness proof when $f < \lfloor \frac{1}{2}(2r+1)(r+1) \rfloor$ in the L_∞ metric

In this section, we prove the correctness (validity, agreement, and integrity) of our proposed BRB protocol when $f < \lfloor \frac{1}{2}(2r+1)(r+1) \rfloor$ in the L_∞ metric. The intuition of our proof is that a valid **certificate** message will be constructed and received by all correct nodes. In addition, in a round, all valid **certificate** messages certify the same message. We consider the worst case where the “important” nodes that are near the source node are faulty. A node near the source node has more shared neighbors with the source node and is more likely to influence the construction of **certificate** messages. As a result, a BRB protocol can handle all cases if it can handle the worst case. To better understand the proof, we first provide the necessary information of Fig. 5 in Table 3.

We first prove the validity.

Lemma 1. *If a source node first calls $\text{r_bcast}(m, \gamma)$ in round γ , then at least one correct node will construct a valid **certificate** for m and broadcast it.*

Proof. Due to the nature of an idealized shared wireless channel, all neighbors of the source node will first receive m in round γ (if the source node is faulty, it may broadcast other messages after m). Then, each correct neighbor will broadcast a **committed** message for m .

Each of the nodes within the yellow region of Fig. 5a (including the boundary) has at least $(2r+1)(r+1)$ shared neighbors with the source node. This is because the node $\mathcal{N}_P$ within the yellow region has the least number of shared neighbors with the source node, i.e., the dark green region in Fig. 5b, which contains $(2r+1)(r+1)$ nodes. The nodes within the yellow region must receive the **committed** messages for m from the correct shared neighbors. As a result, each of them can receive at least $(2r+1)(r+1) - f \geq f + 1$ **committed** messages, which means that any of them can construct a valid **certificate** message. Then, we only need to prove that there is at least one correct node within this yellow region, which is straightforward because the number of nodes within this yellow region is $2r(r+1) + 1 > \lfloor \frac{1}{2}(2r+1)(r+1) \rfloor > f$. Therefore, a valid **certificate** message will be constructed by at least one correct node.

Lemma 2. *If there is a valid **certificate** message for m in round γ , then no valid **certificate** message for m' ($m' \neq m$) in round γ can be constructed.*

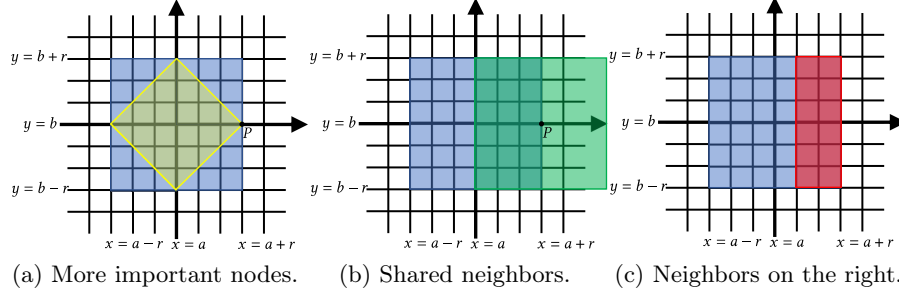


Fig. 5. Correctness proof in an infinite grid under L_∞ metric.

Table 3. The summarized necessary information of Fig. 5.

Region	x -extent and y -extent	Description of nodes	Numbers of nodes
Blue	$a - r \leq x \leq a + r$ $b - r \leq y \leq b + r$	The neighbors of $\mathcal{N}_{(a,b)}$	$(2r + 1)^2$
Yellow	$ x - a + y - b \leq r$	The more important nodes	$(2r + 1)(r + 1)$
Dark green	$a \leq x \leq a + r$ $b - r \leq y \leq b + r$	The shard neighbors of $\mathcal{N}_P$ and $\mathcal{N}_{(a,b)}$	$2r(r + 1) + 1$
Red	$a + 1 \leq x \leq a + r$ $b - r \leq y \leq b + r$	The neighbors of $\mathcal{N}_{(a,b)}$ on the right	$(2r + 1)r$

Proof. We prove this by contradiction. At least one correct node within the neighborhood of the source node broadcasts a **committed** message for m because there is a valid **certificate** message for m . Due to the nature of an idealized wireless channel, all correct nodes within the neighborhood of the source node broadcast **committed** messages for m . Suppose a valid **certificate** message for m' ($m' \neq m$) in round γ is constructed. This implies that at least one correct node within the neighborhood of the source node broadcasts a **committed** message for m' , which leads to a contradiction.

Lemma 3. *If a valid certificate message for m is constructed and broadcast by a correct node, all correct nodes will receive it and reliably deliver m .*

Proof. If a correct node $\mathcal{N}_{(a,b)}$ broadcasts a **certificate** message and at least one correct neighbors in each of four directions (i.e., up $\mathcal{N}_{(*,b+y)}$, down $\mathcal{N}_{(*,b-y)}$, left $\mathcal{N}_{(a-x,*)}$, right $\mathcal{N}_{(a+x,*)}$, $0 < x, y \leq r$) forwards it, then all correct nodes will receive this **certificate** message. Next, we prove that the **certificate** message can be propagated in four directions.

We prove this by contradiction. Suppose a correct node $\mathcal{N}_{(a,b)}$ broadcasts a **certificate** message, but no neighbor on the right forwards this **certificate** message. Then, all these $r(2r + 1)$ neighbors on the right (within the red region of Fig. 5c) should be faulty. However, $r(2r + 1) > f$, which leads to a contradiction. Therefore, if a correct node broadcasts a **certificate** message, at least one correct neighbor on its right will forward this **certificate** message. The same reasoning applies to the other three directions, enabling the message to spread in

four directions. Thus, all correct nodes will receive this **certificate** message for m . By Lemma 2, in the same round, no valid **certificate** message for m' can be constructed. Therefore, all correct nodes will first receive a **certificate** message for m and reliably deliver m .

Theorem 1 (Validity). *If a correct node $\mathcal{N}_i$ calls $\text{r_bcast}(m, \gamma)$, then all correct nodes will output $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$.*

Proof. By Lemma 1, a valid **certificate** message for m in round γ will be constructed and broadcast by a correct node. By Lemma 3, all correct nodes will receive this valid **certificate** message and output $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$.

Next, we prove the agreement.

Theorem 2 (Agreement). *If a correct node outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$, then all other correct nodes will output $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$.*

Proof. Suppose a correct node $\mathcal{N}_i$ outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$, we consider the following two cases:

- *Node $\mathcal{N}_i$ is a neighbor of the source node $\mathcal{N}_j$.* In this case, due to the nature of an idealized shared wireless channel, all neighbors of the source node will output $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$. By Lemma 1, a valid **certificate** message for m in round γ will be constructed and broadcast by a correct node. By Lemma 3, all correct nodes will receive this valid **certificate** message for m and output $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$.
- *Node $\mathcal{N}_i$ is not a neighbor of the source node $\mathcal{N}_j$.* In this case, node $\mathcal{N}_i$ must receive a valid **certificate** message for m . By Lemma 3, all correct nodes will also receive this valid **certificate** message and output $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$.

Therefore, if a correct node outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$, then all other correct nodes will output $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$.

Next, we prove the integrity.

Theorem 3 (Integrity). *For each round γ and node $\mathcal{N}_i$, every correct node outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$ at most once regardless of m .*

Proof. Suppose a correct node $\mathcal{N}_j$ outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$, we consider the following two cases:

- *The node $\mathcal{N}_j$ is the neighbor of the source node $\mathcal{N}_i$.* In this case, $\mathcal{N}_j$ outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$ when it receives a first valid **committed** message from $\mathcal{N}_i$ in round γ .
- *The node $\mathcal{N}_j$ is not the neighbor of the source node $\mathcal{N}_i$.* In this case, $\mathcal{N}_j$ outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$ when it receives a valid **certificate** message for the first time.

Either case has only one rule of outputting. In addition, a node is either a neighbor or a non-neighbor of the source node. Therefore, for each round γ and node $\mathcal{N}_i$, every correct node outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$ at most once regardless of m .

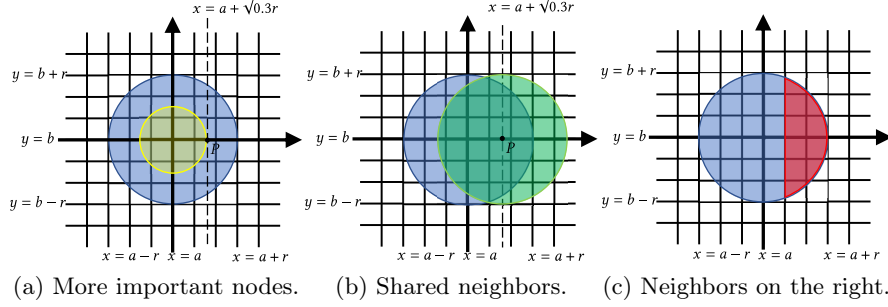


Fig. 6. Correctness proof in an infinite grid under L_2 metric.

Table 4. The summarized necessary information of Fig. 6.

Region	x -extent and y -extent	Description of nodes	Numbers of nodes
Blue	$(x - a)^2 + (y - b)^2 \leq r^2$	The neighbors of $\mathcal{N}_{(a,b)}$	πr^2
Yellow	$(x - a)^2 + (y - b)^2 \leq 0.3r^2$	The more important nodes	$0.3\pi r^2$
Dark green	$(x - a)^2 + (y - b)^2 \leq 0.3r^2$	The shard neighbors of $\mathcal{N}_P$ and $\mathcal{N}_{(a,b)}$	$2r^2 \cos^{-1} \frac{\sqrt{0.3}}{2} - \frac{1}{2}r^2\sqrt{1.11}$
Red	$(x - a)^2 + (y - b)^2 \leq r^2$ $a + 1 \leq x$	The neighbors of $\mathcal{N}_{(a,b)}$ on the right	$0.5\pi r^2 - 2r - 1$

4.2 Correctness proof when $f < \lfloor 0.3\pi r^2 \rfloor$ in the L_2 metric

In this section, we prove the correctness (validity, agreement, and integrity) of our proposed BRB when $f < \lfloor 0.3\pi r^2 \rfloor$ in the L_2 metric. In the L_2 metric, for sufficiently large r , the area can be used as a good approximation for the number of nodes in these regions [4], which serves as the basis for our deductions. The intuition of the proof in the L_2 metric is the same as that in the L_∞ metric. We still consider the worst case where the “important” nodes that are near the source node are faulty. In the L_2 metric, a node near the source node also has more shared neighbors with the source node and is more likely to influence the construction of **certificate** messages. As a result, a BRB protocol can handle all cases if it can handle the worst case. Similarly, to better understand the proof, we provide the necessary information of Fig. 6 in Table 4.

We first prove the validity.

Lemma 4. *If a source node first calls $\text{r_bcast}(m, \gamma)$ in round γ , then at least one correct node will construct a valid **certificate** message for m and broadcast it.*

Proof. Similarly to the proof of Lemma 1 in the L_∞ metric, each correct neighbor of the source node will broadcast a **committed** message for m .

The node $\mathcal{N}_P$ within the yellow region in Fig. 6a has the least number of shared neighbors with the source node. Specifically, their shared neighbors are within the dark green region in Fig. 6b, and their number can be approximated as $N_1 = 2r^2 \cos^{-1} \frac{\sqrt{0.3}}{2} - \frac{1}{2}r^2\sqrt{1.11}$. Thus, each node within the yellow region

has at least N_1 shared neighbors with the source node. When $r \geq 4$, each node within the yellow region can receive at least $N_1 - f \geq f + 1$ committed messages because each of them will receive the committed messages for m from the correct shared neighbors. This means that any of them can construct a valid certificate message for m . Furthermore, there is at least one correct node within this yellow region because $f < \lfloor 0.3\pi r^2 \rfloor$. Therefore, at least one correct node will construct a valid certificate message for m and broadcast it.

Lemma 5. *If there is a valid certificate message for m in round γ , then no valid certificate for m' ($m' \neq m$) in round γ can be constructed.*

Proof. The proof remains identical to that of Lemma 2.

Lemma 6. *If a valid certificate message is constructed and broadcast by a correct node, all correct nodes will receive it and reliably deliver it.*

Proof. Similarly to the proof of Lemma 3, we prove this by proving that a certificate message can be propagated in four directions.

We prove this by contradiction. Suppose that a correct node broadcasts a certificate message, but no correct neighbor on the right forwards this certificate message. Then, all these $r(2r+1)$ neighbors on the right (within the red region of Fig. 6c) should be faulty. However, $(0.5\pi r^2 - 2r - 1) - f \geq 1$ when $r \geq 4$, which leads to a contradiction. Therefore, if a correct node broadcasts a certificate message, at least one correct neighbor on its right will forward this certificate message. The same reasoning applies to the other three directions, enabling the message to spread in four directions. Therefore, if a valid certificate message is constructed and broadcast by a correct node, all correct nodes will receive it. By Lemma 5, in the same round, no valid certificate message for m' can be constructed. Therefore, all correct nodes will first receive a certificate message for m and reliably deliver m .

Theorem 4 (Validity). *If a correct node $\mathcal{N}_i$ calls $\text{r_bcast}(m, \gamma)$, then all correct nodes will output $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$.*

Proof. Based on the Lemma 4, and Lemma 6, the proof of this theorem remains identical to the Theorem 1.

Then, we prove the agreement.

Theorem 5 (Agreement). *If a correct node $\mathcal{N}_i$ outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$, then all other correct nodes will output $\text{r_deliver}(m, \gamma, \mathcal{N}_j)$.*

Proof. With Lemma 4 and Lemma 6, the agreement proof remains identical to that of Theorem 2.

Then, we prove the integrity.

Theorem 6 (Integrity). *For each round γ and node $\mathcal{N}_i$, every correct node outputs $\text{r_deliver}(m, \gamma, \mathcal{N}_i)$ at most once regardless of m .*

Proof. The integrity proof remains identical to that of Theorem 3.

5 Related Work

Byzantine reliable broadcast in wired networks. Byzantine reliable broadcast (BRB) is a fundamental problem in fault-tolerant distributed systems, introduced by Bracha and Toueg [9,8,7]. They show that BRB is achievable in wired asynchronous networks when less than one-third of nodes can be faulty. Then, hundreds of works such as [10,1,2,13] follow this result to dedicate themselves to reducing the communication cost. However, these results cannot be directly adapted to wireless networks because they do not capture the spatially dependent connectivity of wireless networks. Specifically, in a wired network, the connection is in a peer-to-peer manner in which nodes can directly communicate with each other. In contrast, in a wireless network, a node can only communicate directly with the one within its broadcast radius.

Byzantine reliable broadcast in wireless grid networks. In wireless networks, Koo [11] provides the first analysis of BRB in an infinite (or finite-toroidal) wireless grid network under the assumption of no message collisions and address spoofing. He shows how to achieve a BRB that tolerates roughly $\frac{1}{2}r^2$ Byzantine faults in the L_∞ metric. He also proves the impossibility of achieving BRB when more than $\lceil \frac{1}{2}r(2r+1) \rceil$ neighbors of any correct node are Byzantine nodes in the L_∞ metric. Vaikuntanathan [15] shows that the protocol proposed by Koo [11] indeed tolerates $\frac{1}{\sqrt{2}}r^2$ Byzantine faults. Bhandari et al. present a BRB algorithm [4] that matches the impossibility threshold in the L_∞ metric. They also present an approximate threshold $f < 0.23\pi r^2$ in the L_2 metric. Then, they present a simpler characterization [5] of a sufficient condition for BRB to augment [4]. In [12], Koo et al. allow Byzantine nodes to cause a known bounded number of collisions and address spoofing of arbitrary other nodes. They show that without the no-collision assumption, BRB is still achievable and has the same fault-tolerant threshold as the no-collision case of [11,4].

Although current works achieve remarkable results, the established thresholds in wireless grid networks still have a gap with those in wired networks. In this paper, we focus on the no-collision case and dedicate ourselves to exploring the thresholds.

6 Conclusion

In this paper, we focus on the no-collision case of infinite (or finite-toroidal) wireless grid networks to explore the thresholds of the Byzantine reliable broadcast (BRB). We present new thresholds of BRB in the L_∞ and L_2 metrics that are closer to the thresholds in wired networks. Furthermore, we design a BRB protocol that matches these new thresholds with $O(1)$ forwarding frequency. Moreover, we provide detailed derivations of the new thresholds and formal correctness proofs of our BRB. In future work, it will be interesting to know whether tighter thresholds can be obtained. Additionally, we will explore how to adapt this idea for the bounded-collision case.

References

1. Alhaddad, N., Das, S., Duan, S., Ren, L., Varia, M., Xiang, Z., Zhang, H.: Balanced byzantine reliable broadcast with near-optimal communication and improved computation. In: Proceedings of the 2022 ACM Symposium on Principles of Distributed Computing. p. 399–417. PODC’22, Association for Computing Machinery, New York, NY, USA (2022). <https://doi.org/10.1145/3519270.3538475>, <https://doi.org/10.1145/3519270.3538475>
2. Alhaddad, N., Duan, S., Varia, M., Zhang, H.: Succinct erasure coding proof systems. Cryptology ePrint Archive, Paper 2021/1500 (2021), <https://eprint.iacr.org/2021/1500>
3. Bhandari, V., Vaidya, N.H.: Reliable broadcast in wireless networks with probabilistic failures. In: IEEE INFOCOM 2007 - 26th IEEE International Conference on Computer Communications. pp. 715–723 (2007). <https://doi.org/10.1109/INFCOM.2007.89>
4. Bhandari, V., Vaidya, N.H.: On reliable broadcast in a radio network. In: Proceedings of the Twenty-Fourth Annual ACM Symposium on Principles of Distributed Computing. p. 138–147. PODC ’05, Association for Computing Machinery, New York, NY, USA (2005). <https://doi.org/10.1145/1073814.1073841>, <https://doi.org/10.1145/1073814.1073841>
5. Bhandari, V., Vaidya, N.H.: On reliable broadcast in a radio network: A simplified characterization. CSL, UIUC, Tech. Rep (2005)
6. Boneh, D., Lynn, B., Shacham, H.: Short signatures from the weil pairing. In: Boyd, C. (ed.) Advances in Cryptology — ASIACRYPT 2001. pp. 514–532. Springer Berlin Heidelberg, Berlin, Heidelberg (2001)
7. Bracha, G.: Asynchronous byzantine agreement protocols. Information and Computation **75**(2), 130–143 (1987). [https://doi.org/https://doi.org/10.1016/0890-5401\(87\)90054-X](https://doi.org/https://doi.org/10.1016/0890-5401(87)90054-X), <https://www.sciencedirect.com/science/article/pii/S089054018790054X>
8. Bracha, G., Toueg, S.: Resilient consensus protocols. In: Proceedings of the Second Annual ACM Symposium on Principles of Distributed Computing. p. 12–26. PODC ’83, Association for Computing Machinery, New York, NY, USA (1983). <https://doi.org/10.1145/800221.806706>, <https://doi.org/10.1145/800221.806706>
9. Bracha, G., Toueg, S.: Asynchronous consensus and broadcast protocols. J. ACM **32**(4), 824–840 (Oct 1985). <https://doi.org/10.1145/4221.214134>, <https://doi.org/10.1145/4221.214134>
10. Das, S., Xiang, Z., Ren, L.: Asynchronous data dissemination and its applications. In: Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security. p. 2705–2721. CCS ’21, Association for Computing Machinery, New York, NY, USA (2021). <https://doi.org/10.1145/3460120.3484808>, <https://doi.org/10.1145/3460120.3484808>
11. Koo, C.Y.: Broadcast in radio networks tolerating byzantine adversarial behavior. In: Proceedings of the Twenty-Third Annual ACM Symposium on Principles of Distributed Computing. p. 275–282. PODC ’04, Association for Computing Machinery, New York, NY, USA (2004). <https://doi.org/10.1145/1011767.1011807>, <https://doi.org/10.1145/1011767.1011807>
12. Koo, C.Y., Bhandari, V., Katz, J., Vaidya, N.H.: Reliable broadcast in radio networks: the bounded collision case. In: Proceedings of the Twenty-Fifth Annual ACM Symposium on Principles of Distributed Computing. p. 258–264. PODC ’06, Association for Computing Machinery, New York, NY, USA (2006). <https://doi.org/10.1145/1146381.1146420>, <https://doi.org/10.1145/1146381.1146420>

13. Nayak, K., Ren, L., Shi, E., Vaidya, N.H., Xiang, Z.: Improved extension protocols for byzantine broadcast and agreement. CoRR **abs/2002.11321** (2020), <https://arxiv.org/abs/2002.11321>
14. Schnorr, C.P.: Efficient signature generation by smart cards. *Journal of cryptology* **4**, 161–174 (1991)
15. Vaikuntanathan, V.: Brief announcement: broadcast in radio networks in the presence of byzantine adversaries. In: *Proceedings of the Twenty-Fourth Annual ACM Symposium on Principles of Distributed Computing*. p. 167. PODC '05, Association for Computing Machinery, New York, NY, USA (2005). <https://doi.org/10.1145/1073814.1073844>, <https://doi.org/10.1145/1073814.1073844>